# Protik Dev

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## Summary

Full-stack software engineer (TypeScript, Next.js, Go) with ∼3 years of corporate experience and 2+ years of freelance experience, building products from UI to production infrastructure.

## Education

**BGC Trust University**
*B.Sc. in Computer Science and Engineering* — *Chattogram, Bangladesh*

## Professional Experience

**Devspace** — **Mar 2026 – Aug 2026**
*Software Engineer*

- Built full-stack React and TypeScript features on an **agentic AI pipeline** (Mastra AI, Convex, Vercel AI SDK).
- Built a **dynamic tool registry**: agent tools are stored in the database and discovered at runtime, not hardcoded.
- Contributed to CI/CD pipeline configuration and deployment automation.
- **Tech:** *React, TypeScript, Convex, Vercel AI SDK, GitHub Actions, Codex*

**Osilion** — **Jul 2025 – Feb 2026**
*Full-Stack Engineer*

- Served as one of the **primary engineers** on a client's internal tool for improving employee productivity.
- Worked on an intelligent recruitment platform, contributing to its technical architecture, database design, and component design patterns.
- Mentored a junior developer through regular code reviews.
- **Tech:** *Next.js, TypeScript, Tailwind, shadcn/ui, Zustand, Appfarm (low-code), C#, Azure, SQL Server*

**Hone** — **Aug 2025 – Sep 2025**
*Frontend Engineer*

- Implemented a **Smart Scaling system** that automatically adapts the desktop app to smaller screens.
- Owned the review and download pages end to end; found and fixed a small-screen layout issue with a CSS grid pattern, validated through manual cross-device testing.
- Improved code splitting and lazy loading to cut initial load weight, and fixed a keyboard focus-visibility issue.
- **Tech:** *React, TypeScript, Electron, Tailwind, shadcn/ui, Claude Code*

**Hello World Communications** — **Sep 2024 – Jul 2025**
*Full-Stack Engineer*

- **Led the development flow end to end** across multiple concurrent client projects, conducting code reviews and making key architectural decisions.
- Reduced client closing time by **roughly 50%** through better scoping and client-to-team delivery alignment.
- Mentored a junior developer across multiple client projects.
- **Tech:** *Next.js, TypeScript, Go, shadcn/ui, TanStack Query, Zustand, Drizzle ORM, Appwrite, Firebase, Docker, SQL Server*

**Bilsida** — **Oct 2023 – Aug 2024**
*Full-Stack Engineer*

- Built a Swedish car marketplace **from scratch across the full stack**; it went on to attract **two of Sweden's largest car dealers**.
- Owned backend logic, database schema design, image optimization, and SEO.
- Mentored a junior developer in building a chat system.
- **Tech:** *Next.js, TypeScript, Tailwind, shadcn/ui, Appwrite, Docker, Context API*

**Fiverr** — **Jul 2021 – Oct 2023**
*Freelance Full-Stack Developer* — *Remote*

- Delivered full-stack web applications and APIs for **20+ clients**, covering frontend, backend, and deployment.

## Backend & Systems Projects

**Spud** Live | Source

- Built and deployed to production a Discord bot for small-team task coordination with Gemini-powered features: **duplicate-task detection** that blocks redundant work at claim time, and a **scope-drift detector** that diffs a git branch against its natural-language task description and pings the owner in-channel when work strays outside the claimed scope.
- Engineered a **GitHub App integration** end to end: short-lived installation tokens, an app-level **Webhook Pipeline**, per-project **Rate Limiting**, and automatic task closure on PR merge.
- Added a **Role-based Permission Model** (team lead vs. server admin) enforced in application code.
- Set up **production observability**: structured JSON logging shipped to **Grafana Loki**, plus **Synthetic Monitoring** with uptime alerting; containerized with Docker on a hosted Turso (libSQL) database.
- **Tech:** *TypeScript, Bun, discord.js, GitHub Apps API, GitHub Actions, Gemini, Vercel AI SDK, SQLite/Turso, Grafana (Loki + Synthetic Monitoring), Docker*

**Result Lookup** Source

- Designed and built a high-throughput exam result publishing system **serving 2M results**.
- Reached ∼**32k RPS** peak with **p99 under 105 ms at 1,000 concurrent connections**, backed by a PostgreSQL fallback with connection limiting.
- **Tech:** *Go, PostgreSQL, Redis, pgx/v5, Docker, golang-migrate, slog, miniredis*

**URL Health Checker** Source

- Built an async job queue with separate API and worker binaries; workers claim jobs via `SELECT ... FOR UPDATE SKIP LOCKED`, with **exponential backoff written in SQL**, atomic with status updates.
- Recovered stale jobs with a staleness threshold instead of a heartbeat, avoiding a race on bounded-duration jobs; graceful shutdown via `signal.NotifyContext`, with integration tests against real PostgreSQL.
- **Tech:** *Go, PostgreSQL, pgx/v5, Docker, golang-migrate, slog*

## Full-Stack Projects

**Narrative Guard** Live | Source

- Built a workspace-based brand governance platform with **role-based ownership enforcement** and **per-brand RAG namespaces** for workspace isolation.
- Built the audit flow end to end (score, verdict, findings, and an AI rewrite) in a responsive Next.js UI, with invite tokens stored as hashes and scheduled Convex actions handling async audit processing.
- **Tech:** *Next.js, TypeScript, Convex, Convex Auth, Google Gemini, RAG, shadcn/ui, Vitest, React Testing Library*

**Roadmap App** Live | Source

- Built a full-stack app with SSR, authentication, a REST API, and upvoting and commenting.
- Covered with tests using Jest and React Testing Library.
- **Tech:** *Next.js, TypeScript, Drizzle ORM, PostgreSQL, Vercel, Jest, React Testing Library*

## Technical Skills

**Languages**: TypeScript, JavaScript, Go, SQL
**Frameworks**: React.js, Next.js (SSR, ISR, SSG), Node.js, Express.js
**Frontend**: Tailwind CSS, shadcn/ui, Framer Motion
**State Management**: Context API, Zustand, React Query
**Databases**: PostgreSQL, MongoDB, MySQL, SQLite, Redis, Drizzle ORM
**AI & Agents**: Vercel AI SDK, Google Gemini, RAG, Claude Code, Codex
**DevOps/Infra**: Docker, GitHub Actions, Vercel, Firebase, Appwrite, Convex, AWS (basic)
**Testing & Tools**: Git, Jest, Vitest, React Testing Library
**Soft Skills**: Communication, Ownership, Leadership, Collaboration, Analytical Thinking, Friendly & Approachable